Biological molecules

Original exam-style questions · 20 questions

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Question 1

core · 1 mark · 0610_T04_P1_20261013_Q1

Which chemical element is found in proteins but not in carbohydrates or fats?

A carbon

B nitrogen

C oxygen

D hydrogen

Question 2

core · 1 mark · 0610_T04_P1_20261013_Q2

Which row shows a large molecule and the smaller molecules it is made from?

	large molecule	**smaller molecules**
A	cellulose	amino acids
B	glycogen	fatty acids and glycerol
C	protein	glucose
D	starch	glucose

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Question 3

core · 1 mark · 0610_T04_P1_20261013_Q3

Samira tests a sample of a breakfast cereal for three nutrients.

- After adding iodine solution, the sample stays orange-brown.
- After heating with Benedict's solution, the mixture stays blue.
- After adding biuret solution, the mixture turns purple.

Which of the nutrients tested is present in the cereal?

A protein only

B protein and reducing sugar

C starch and protein

D starch only

Question 4

core · 1 mark · 0610_T04_P1_20261013_Q4

Which row correctly describes a food test and its positive result?

	test	**method**	**positive result**
A	iodine solution	add a few drops to the sample	orange-brown colour
B	Benedict's solution	add to the sample and leave at room temperature	brick-red colour
C	ethanol emulsion	shake the sample with ethanol, then pour the ethanol into water	milky white emulsion
D	DCPIP	add the sample drop by drop to DCPIP	colourless liquid turns blue

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Question 5

core · 1 mark · 0610_T04_P1_20261013_Q5

Four drinks were tested for vitamin C. Each drink was added drop by drop to 1 cm^3 of DCPIP solution until the blue colour disappeared.

drink	number of drops needed
1	18
2	6
3	25
4	11

Which drink contains the highest concentration of vitamin C?

A drink 2

B drink 4

C drink 1

D drink 3

Question 6

core · 1 mark · 0610_T04_P1_20261013_Q6

Which statement about starch, glycogen and cellulose is correct?

A They are all made from amino acids.

B They are all made from glucose.

C They all contain nitrogen.

D Only starch contains carbon.

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Question 7

core · 1 mark · 0610_T04_P1_20261013_Q7

Four sugar solutions, W, X, Y and Z, were each heated with the same volume of Benedict's solution for the same time. The colours at the end were:

solution	colour after heating
W	green
X	brick red
Y	blue
Z	yellow

Which list shows the solutions in order of increasing concentration of reducing sugar?

A X, Z, W, Y

B W, Y, Z, X

C Y, Z, W, X

D Y, W, Z, X

Question 8

supplement · 1 mark · 0610_T04_P1_20261013_Q8

Which statement describes the structure of a DNA molecule?

A one strand of bases coiled into a helix

B two strands held together by bonds between glucose molecules

C two strands coiled into a double helix, held together by bonds between pairs of bases

D two strands in which base A pairs with base C, and base T pairs with base G

Question 9

supplement · 1 mark · 0610_T04_P1_20261013_Q9

In a sample of DNA, 24% of the bases are C.

What percentage of the bases in the sample are A?

A 24%

B 26%

C 48%

D 52%

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Question 10

core · 1 mark · 0610_T04_P1_20261013_Q10

The white powder in a jar contains only one type of biological molecule. The molecule is a carbohydrate, a fat or a protein.

Chemical analysis shows that the powder contains carbon, hydrogen, oxygen and nitrogen.

Which food test result is expected for this powder?

A Biuret solution turns purple.

B Iodine solution turns blue-black.

C The ethanol emulsion test gives a milky white emulsion.

D Benedict's solution turns brick red after heating.

Question 11

core · 5 marks · 0610_T04_P1_20261013_Q11

Leila eats a breakfast of oats and milk. Oats are rich in starch. Milk contains protein and fat.

(a) State the three chemical elements that are found in both carbohydrates and fats. [1]

(b) Starch is made from glucose. State the name of one other large carbohydrate molecule that is made from glucose. [1]

(c) State the name of the smaller molecules that join together to make proteins. [1]

(d) Fats are made from two types of smaller molecule. State the names of both types. [2]

Question 12

core · 4 marks · 0610_T04_P1_20261013_Q12

Tomas wants to find out whether a sample of crushed rice contains starch.

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(a) State the name of the reagent Tomas should add to the rice. [1]

(b) State the colour Tomas will see if starch is present. [1]

(c) State the colour Tomas will see if starch is not present. [1]

(d) Starch is a large molecule. State the name of the smaller molecule that starch is made from. [1]

Question 13

core · 6 marks · 0610_T04_P1_20261013_Q13

Kofi is given a sample of crushed peanuts. He uses the ethanol emulsion test to find out whether the peanuts contain fat.

(a) Describe how Kofi should carry out the ethanol emulsion test on the crushed peanuts. [2]

(b) State the result Kofi will see if fat is present. [1]

(c) Kofi repeats the test using distilled water in place of the crushed peanuts. Suggest why he does this. [1]

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(d) Kofi also wants to find out whether the peanuts contain protein. Describe how he should test for protein and state the result if protein is present. [2]

Question 14

core · 5 marks · 0610_T04_P1_20261013_Q14

Priya compares the vitamin C content of fresh orange juice and orange juice that has been boiled and then cooled. She adds each juice slowly from a syringe to 1.0 cm³ of DCPIP solution until the DCPIP changes colour. She does this three times for each juice.

juice	trial 1 volume / cm³	trial 2 volume / cm³	trial 3 volume / cm³	mean volume / cm³
fresh	1.2	1.4	1.3	1.3
boiled	2.5	2.7	2.9	

(a) State the colour change of DCPIP when enough vitamin C is added. [1]

(b) Calculate the mean volume of boiled juice needed. [1]

(c) State which juice contained the higher concentration of vitamin C. Explain your answer using the data in the table. [2]

(d) Suggest one variable that Priya should keep the same in every trial. [1]

Question 15

supplement · 4 marks · 0610_T04_P1_20261013_Q15

DNA is found in the nucleus of a cell.

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(a) Describe the structure of a DNA molecule. [3]

(b) One student writes: "In DNA, base A always pairs with base G."

State a corrected version of this sentence. [1]

Question 16

supplement · 4 marks · 0610_T04_P1_20261013_Q16

Scientists analysed the DNA from the cells of a wheat plant. They found that 27% of the bases in the DNA were A.

(a) State the percentage of the bases that were T. Explain your answer. [2]

(b) Calculate the percentage of the bases that were G. Show your working. [2]

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Question 17

core · 6 marks · 0610_T04_P1_20261013_Q17

There are four unlabelled tubes, J, K, L and M. One tube contains glucose solution, one contains starch suspension, one contains protein solution and one contains distilled water. A student tests a sample from each tube.

tube	iodine solution	Benedict's solution, after heating	biuret solution
J	orange-brown	blue	purple
K	blue-black	blue	blue
L	orange-brown	brick red	blue
M	orange-brown	blue	blue

(a) Identify the contents of each tube, J, K, L and M. [4]

(b) The student then tests a fifth liquid, N. The iodine solution turns blue-black, the Benedict's solution turns orange after heating and the biuret solution stays blue.

State which nutrients liquid N contains. [2]

Question 18

core · 6 marks · 0610_T04_P1_20261013_Q18

Nadia mixes a starch suspension with a liquid extracted from germinating barley seeds. She keeps the mixture at 35 °C. She tests samples of the mixture with iodine solution and with Benedict's solution at the start and after 30 minutes.

time / min	iodine solution	Benedict's solution, after heating
0	blue-black	blue
30	orange-brown	orange

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(a) Explain the change in the result of the iodine test. [2]

(b) Explain the change in the result of the Benedict's test. [2]

(c) Nadia repeats the experiment using a suspension of cellulose instead of starch. After 30 minutes the Benedict's test still stays blue.

Suggest what this result shows about the liquid from the barley seeds. [1]

(d) Nadia also heats a sample of the barley seed liquid on its own with Benedict's solution. It stays blue.

Suggest why she carries out this test. [1]

Question 19

supplement · 6 marks · 0610_T04_P1_20261013_Q19

This section of a DNA molecule has 10 bases on each strand, 20 bases in total. The order of bases on strand 1 is shown.

Strand 1: C G A T T A C G G A

(a) State the order of bases on strand 2. [1]

(b) Calculate the percentage of the 20 bases in this section that are C. Show your working. [2]

(c) When this DNA is heated to a high temperature, the two strands separate from each other, but each strand stays in one piece.

Suggest which bonds are broken. Explain your answer. [2]

(d) Explain why the number of A bases in the whole section is equal to the number of T bases. [1]

Question 20

core · 6 marks · 0610_T04_P1_20261013_Q20

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Diego tests extracts from bananas at three stages of ripeness for starch and for reducing sugar. He uses the same mass of banana to make each extract.

stage of banana	iodine solution	Benedict's solution, after heating
unripe (green skin)	blue-black	green
ripe (yellow skin)	blue-black	orange
overripe (brown skin)	orange-brown	brick red

(a) Describe how Diego should carry out the Benedict's test on each extract. [2]

(b) Describe how the starch and reducing sugar content of the banana change as it ripens. [2]

(c) State two variables, other than the mass of banana, that Diego should keep the same so that the Benedict's results can be compared fairly. [2]

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